

Port Scan Investigation — port_scan.pcap

Objective

Identify and document evidence of port scanning activity in the provided PCAP capture: determine the scanning host, discovered hosts, and key network identifiers.

Tools & Methods

- Wireshark (PCAP inspection, display filters)
- Observed TCP flags and conversation patterns (SYN floods, SYN→RST responses)

Summary of Steps Taken

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.42.42.253	10.42.42.58	TCP	76	48184 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=329940 TSecr=0 WS=64
2	0.000021	10.42.42.58	10.42.42.253	TCP	60	88 → 48184 [RST] Seq=1 Win=0 Len=0
3	0.007594	10.42.42.253	10.42.42.56	TCP	74	58056 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=330092 TSecr=0 WS=64
4	0.007596	10.42.42.253	10.42.42.25	TCP	74	48021 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=330092 TSecr=0 WS=64
5	0.007597	10.42.42.253	10.42.42.253	TCP	60	→ 59106 [RST] Seq=1 Win=0 Len=0
6	0.007769	10.42.42.253	10.42.42.253	TCP	60	88 → 48921 [RST] Seq=1 Ack=1 Win=0 Len=0
7	0.012799	10.42.42.253	10.42.42.58	TCP	74	38232 → 554 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
8	0.012793	10.42.42.253	10.42.42.56	TCP	74	43771 → 554 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
9	0.013112	10.42.42.58	10.42.42.253	TCP	60	554 → 43771 [RST] Seq=1 Ack=1 Win=0 Len=0
10	0.013968	10.42.42.253	10.42.42.25	TCP	74	58085 → 554 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
11	0.013979	10.42.42.253	10.42.42.58	TCP	74	35168 → 389 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
12	0.013291	10.42.42.253	10.42.42.56	TCP	74	43214 → 389 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
13	0.013263	10.42.42.25	10.42.42.253	TCP	60	554 → 58085 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
14	0.013267	10.42.42.56	10.42.42.253	TCP	60	389 → 43514 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
15	0.013322	10.42.42.253	10.42.42.25	TCP	74	49945 → 389 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
16	0.013428	10.42.42.253	10.42.42.58	TCP	74	37966 → 256 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
17	0.013457	10.42.42.253	10.42.42.56	TCP	74	33239 → 256 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
18	0.013459	10.42.42.58	10.42.42.253	TCP	60	554 → 38232 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
19	0.013554	10.42.42.58	10.42.42.253	TCP	60	389 → 35168 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
20	0.013522	10.42.42.56	10.42.42.253	TCP	60	256 → 33239 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
21	0.013529	10.42.42.25	10.42.42.253	TCP	60	389 → 49945 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	0.013588	10.42.42.253	10.42.42.253	TCP	60	256 → 37966 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
23	0.013981	10.42.42.253	10.42.42.58	TCP	74	35682 → 23 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
24	0.013983	10.42.42.253	10.42.42.56	TCP	74	68559 → 23 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
25	0.014075	10.42.42.56	10.42.42.253	TCP	60	23 → 68559 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	0.014096	10.42.42.58	10.42.42.253	TCP	60	23 → 39682 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	0.014222	10.42.42.253	10.42.42.25	TCP	74	68419 → 256 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
28	0.014238	10.42.42.253	10.42.42.58	TCP	74	46561 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
29	0.014308	10.42.42.253	10.42.42.56	TCP	74	59981 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
30	0.014432	10.42.42.56	10.42.42.253	TCP	60	88 → 59981 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
31	0.014434	10.42.42.25	10.42.42.253	TCP	60	256 → 68419 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
32	0.014495	10.42.42.253	10.42.42.25	TCP	74	46672 → 23 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
33	0.014548	10.42.42.253	10.42.42.58	TCP	74	59786 → 23 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
34	0.014617	10.42.42.25	10.42.42.253	TCP	60	23 → 46672 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
35	0.014628	10.42.42.253	10.42.42.56	TCP	74	58953 → 22 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
36	0.014742	10.42.42.25	10.42.42.253	TCP	60	22 → 58953 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37	0.014757	10.42.42.58	10.42.42.253	TCP	60	88 → 46561 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
38	0.014776	10.42.42.58	10.42.42.253	TCP	60	22 → 59786 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
39	0.014799	10.42.42.253	10.42.42.25	TCP	74	43018 → 88 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
40	0.014882	10.42.42.253	10.42.42.58	TCP	74	33697 → 21 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
41	0.014910	10.42.42.25	10.42.42.56	TCP	74	41881 → 21 [SYN] Seq=0 Win=5848 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64

- Loaded port_scan.pcap in Wireshark.
- Inspected traffic volume and TCP flags to spot scanning behaviour: large number of SYN packets to many destination ports followed by RST responses.
- Filtered and grouped by source IP to find the primary scanner.
- Examined destination IPs that received the scan traffic to identify hosts found by the scanner.
- Queried Ethernet frames associated with discovered hosts to extract MAC addresses (for the Apple system).
- Used an IP filter to separate hosts and identify the detected Windows system.

Q1 — What is the IP address scanning the environment?

Evidence: The majority of outbound SYN traffic in the capture originates from 10.42.42.253 hitting many destination ports — classic port-scan behavior.

Answer: 10.42.42.253

Q2 — What is the IP address found as a result of the scan?

Evidence: This IP received multiple scan attempts and replied to the probing connections.

Answer: 10.42.42.50

Q3 — What is the MAC address of the Apple system it finds?

Evidence: The Ethernet frames for the Apple host contain this MAC; correlating IP↔MAC in ARP/packet headers identifies the Apple device.

```
[Time delta from previous captured frame: 0.000090000 seconds]
[Time delta from previous displayed frame: 0.000090000 seconds]
[Time since reference or first frame: 0.607769000 seconds]
Frame Number: 6
Frame Length: 60 bytes (480 bits)
Capture Length: 60 bytes (480 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
[Coloring Rule Name: TCP RST]
[Coloring Rule String: tcp.flags.reset eq 1]
▼ Ethernet II, Src: Apple_92:6e:dc (00:16:cb:92:6e:dc), Dst: QuantaComput_82:1f:4a (00:23:8b:82:1f:4a)
  ▼ Destination: QuantaComput_82:1f:4a (00:23:8b:82:1f:4a)
    Address: QuantaComput_82:1f:4a (00:23:8b:82:1f:4a)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  ▼ Source: Apple_92:6e:dc (00:16:cb:92:6e:dc)
    Address: Apple_92:6e:dc (00:16:cb:92:6e:dc)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  Padding: 000000000000
  ▼ Internet Protocol Version 4, Src: 10.42.42.25, Dst: 10.42.42.253
    0100 .... = Version: 4
```

Answer: 00:16:cb:92:6e:dc

Q4 — What is the IP address of the detected Windows system?

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.42.42.253	10.42.42.50	TCP	74	46104 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3299940 TSecr=0 WS=64
2	0.000231	10.42.42.253	10.42.42.253	TCP	60	80 → 46104 [RST, ACK] Seq=0 Ack=1 Win=0 Len=0
3	0.607594	10.42.42.253	10.42.42.56	TCP	74	59856 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300092 TSecr=0 WS=64
4	0.607679	10.42.42.56	10.42.42.253	TCP	60	80 → 59856 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	0.812790	10.42.42.253	10.42.42.50	TCP	74	38232 → 554 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
8	0.812793	10.42.42.253	10.42.42.56	TCP	74	43771 → 554 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
9	0.812877	10.42.42.56	10.42.42.253	TCP	60	554 → 43771 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
11	0.813070	10.42.42.253	10.42.42.50	TCP	74	35168 → 389 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
12	0.813201	10.42.42.253	10.42.42.56	TCP	74	43514 → 389 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
14	0.813267	10.42.42.253	10.42.42.253	TCP	60	310 → 43514 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
16	0.813429	10.42.42.253	10.42.42.50	TCP	74	37866 → 256 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
17	0.813457	10.42.42.253	10.42.42.56	TCP	74	33239 → 256 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
18	0.813459	10.42.42.50	10.42.42.253	TCP	60	554 → 38232 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
19	0.813514	10.42.42.50	10.42.42.253	TCP	60	389 → 35168 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
20	0.813522	10.42.42.56	10.42.42.253	TCP	60	256 → 33239 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	0.813588	10.42.42.50	10.42.42.253	TCP	60	256 → 37866 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
23	0.813981	10.42.42.253	10.42.42.50	TCP	74	39682 → 23 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
24	0.813983	10.42.42.253	10.42.42.56	TCP	74	68559 → 23 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300144 TSecr=0 WS=64
25	0.814075	10.42.42.56	10.42.42.253	TCP	60	23 → 68559 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	0.814096	10.42.42.50	10.42.42.253	TCP	60	23 → 39682 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
28	0.814230	10.42.42.253	10.42.42.50	TCP	74	46561 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
29	0.814369	10.42.42.253	10.42.42.56	TCP	74	59941 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
30	0.814432	10.42.42.56	10.42.42.253	TCP	60	80 → 59941 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	0.814540	10.42.42.253	10.42.42.50	TCP	74	59706 → 22 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
35	0.814620	10.42.42.253	10.42.42.56	TCP	74	58953 → 22 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
36	0.814712	10.42.42.50	10.42.42.253	TCP	60	22 → 50953 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37	0.814757	10.42.42.50	10.42.42.253	TCP	60	80 → 46561 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
38	0.814776	10.42.42.50	10.42.42.253	TCP	60	22 → 59706 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
40	0.814882	10.42.42.253	10.42.42.50	TCP	74	33697 → 21 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
41	0.814975	10.42.42.253	10.42.42.56	TCP	74	51807 → 21 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
42	0.815038	10.42.42.56	10.42.42.253	TCP	60	21 → 51807 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
45	0.815236	10.42.42.253	10.42.42.50	TCP	74	51331 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
46	0.815351	10.42.42.253	10.42.42.56	TCP	74	46114 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
47	0.815415	10.42.42.50	10.42.42.253	TCP	60	21 → 51331 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
48	0.815417	10.42.42.56	10.42.42.253	TCP	60	443 → 46114 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
50	0.815421	10.42.42.50	10.42.42.253	TCP	60	443 → 51331 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
52	0.815541	10.42.42.253	10.42.42.50	TCP	74	36834 → 3389 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
54	0.815634	10.42.42.253	10.42.42.56	TCP	74	50121 → 3389 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
55	0.815722	10.42.42.56	10.42.42.253	TCP	60	3389 → 50121 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
57	0.815892	10.42.42.253	10.42.42.50	TCP	74	34135 → 113 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
58	0.815999	10.42.42.253	10.42.42.56	TCP	74	50813 → 113 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=3300145 TSecr=0 WS=64
59	0.816073	10.42.42.50	10.42.42.253	TCP	60	113 → 34135 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Evidence: After excluding the Mac host (filter used: ip.addr != 10.42.42.25), the remaining target responding to scans is 10.42.42.50, consistent with a Windows host.

Answer: 10.42.42.50

Observational Note (scan behavior):

By inspecting the TCP exchanges we observed many attempted three-way handshakes where the scanner initiated SYN's and many responses resulted in RST — indicating probes across both common and uncommon ports rather than normal service traffic. This is strong evidence of active port-scanning reconnaissance.

Recommendations

- Block or rate-limit traffic from 10.42.42.253 until the source is investigated.
- Ensure host-based firewalls on 10.42.42.50 and 10.42.42.25 are hardened and logging is enabled.
- Correlate this scan with IDS/IPS logs and any external threat intelligence to determine origin (internal vs external).
- Consider adding detection rules for high-rate SYN probes and automated scans.